FastSpeech 2: English and Telugu Speech Synthesis

Technical Report

Prepared: 07 November 2026

1. Project Overview

This project implements and evaluates a FastSpeech 2 text-to-speech pipeline for English and Telugu. The experiments use the specified Svarah English dataset and IndicTTS Telugu dataset. The workflow includes data preparation, alignment or duration preparation, acoustic feature extraction, model training, inference and analysis of duration, pitch and energy predictions.

2. Datasets and Preparation

English: ai4bharat/Svarah. The verified experiment used 2,662 training and 501 validation feature files, with no missing alignments among the selected feature IDs. The dataset readiness audit recorded approximately 5.00 hours of training audio and 1.00 hour of validation audio, with no train/validation ID overlap.

Telugu: SPRINGLab/IndicTTS_Telugu. The prepared split contained approximately 5.01 hours of training data and 1.00 hour of validation data. The experiment used MMS forced alignment with Uroman romanization and extracted mel, pitch, energy and duration features. The prepared features were chunked for model input.

3. Model and Configuration

Both experiments use a compact FastSpeech 2 configuration with Transformer encoder and decoder layers, a variance adaptor for duration, pitch and energy, and an 80-channel mel-spectrogram representation. The experiments use a 22,050 Hz audio sampling rate and a HiFi-GAN vocoder. Language-specific preprocessing and model configurations are retained in the project folders.

4. English Experiment

The English experiment has a best checkpoint at step 5,000 with recorded best validation loss 3.6294. The latest checkpoint is at step 6,000. Five test utterances were synthesized, and the project contains the corresponding prediction files and duration, pitch and energy plots, alongside training history and loss plots.

5. Telugu Experiment and Limitation

Telugu training reached checkpoint step 600. Five test utterances were synthesized from the recovery checkpoint, and five prediction files plus 15 duration, pitch and energy plots were generated. Inference completed successfully using the universal HiFi-GAN vocoder.

Training limitation: A checkpoint comparison found that trainable model parameters and Adam optimizer state remained unchanged after step 300, while BatchNorm running statistics continued to change. The exact cause has not been established. Therefore, checkpoint 600 is reported as the recovery inference checkpoint, not as evidence that Telugu optimization converged. Telugu audio quality should be assessed separately and results interpreted cautiously.

6. Inference and Analysis

The inference outputs include five utterances per language. For each utterance, the pipeline records predicted mel features, duration predictions, pitch and energy outputs, and generated waveform audio. Variance-adaptor plots are provided to support inspection of duration, pitch and energy behavior.

7. Deliverables

The submission package includes language-specific experiment folders, available checkpoints and configurations, training logs, inference audio, prediction artifacts, plots, this report, README and dependency list. The accompanying audit JSON records which expected artifacts were detected at packaging time.

8. Reproducibility Notes

Use the language-specific configuration files and checkpoints. The prepared feature files and alignments are retained to avoid repeating expensive preprocessing. The Python dependencies are listed in requirements.txt; exact compatibility may depend on the Colab runtime and installed PyTorch/CUDA versions.

Packaging audit

Artifact check	Detected
English best checkpoint	PASS
English latest checkpoint	PASS
English training history	PASS
English inference summary	NOT FOUND / CHECK
Telugu recovery checkpoint	NOT FOUND / CHECK
Telugu recovery inference summary	PASS
English audio outputs	PASS
Telugu recovery audio outputs	PASS
English variance-adaptor plots	PASS
Telugu variance-adaptor plots	PASS